



Aerated Facultative Lagoons Performance Bioaugmented With *Bacillus Thuringiensis* Immobilized On Rice Husks for Treatment of Cellulose Effluents

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Received: 30 March 2026 / Accepted: 21 September 2026
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Abstract This study evaluated the performance of bench and pilot-scale facultative aerated lagoons (FAL) treating Kraft pulp effluents subjected to organic load shocks, with bioaugmentation of *Bacillus thuringiensis* NR114581.1 immobilized on rice husks. The effluents from the treatments were characterized in terms of chemical oxygen demand (COD), biochemical oxygen demand (BOD₅), specific lignin compounds, color, and turbidity. Batch tests were performed to optimize the bioproduct dosage and evaluate its removal performance. During bioaugmentation in the bench-scale FAL, the average removal reached 72% for BOD₅ and 36% for COD, while the removal of specific compounds was low or zero. In the pilot system, a consistent performance under shock conditions was observed, with removals of 60% for BOD₅ and 54% for COD, with limited removal of specific compounds. The biotransformation of organic matter was evaluated using fluorescence excitation-emission

matrix (FEEM) analysis, revealing reductions and increases related to fluorophores associated with protein-like and humic compounds. The microfauna presented typical ecological succession, with predominance of crawling ciliates and metazoans in both systems. Variations in abundance indicated the sensitivity of the systems when subjected to disturbances. The immobilization of bacteria on rice husks with biofilm formation was associated with system performance under the evaluated conditions. The results support the use of FEEM analysis and microfauna monitoring as diagnostic tools for evaluating the performance and resilience of FAL applied to load shocks and bioaugmentation.

Highlights

- Bioaugmentation with *B. thuringiensis* immobilized on rice husks resulted in increased BOD₅ removal in bench-scale and pilot-scale facultative aerated lagoons.
- The pilot-scale facultative aerated lagoon showed greater stability and recovery in the face of organic load shocks than the bench-scale lagoon.
- FEEM analysis revealed changes in protein-like and humic fluorophores during organic load shocks and bioaugmentation.
- The microfauna showed typical ecological succession, with an increase in creeping ciliates and metazoans after bioaugmentation.

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- Immobilization of bacteria on rice husks allowed biofilm formation, suggesting a promising treatment optimization strategy.

Keywords Biological treatment · Recalcitrant compounds · Kraft effluent · Biofilm

1 Introduction

The Kraft process is widely adopted on a global scale. In this process, pulping is carried out with the addition of NaOH and Na₂S under high temperatures and pressures, requiring between 25 and 225 m³ of water per ton of cellulose produced (Neves et al., 2017; Ramos et al., 2022). This industry is also considered to be a major pollutant, generating an effluent composed of a mixture of more than 700 organic and inorganic compounds. Even when producing unbleached cellulose, the disposal of these residues significantly changes the water physicochemical characteristics, reducing the levels of dissolved oxygen, negatively impacting the water quality and aquatic ecosystems (Esmaeeli et al., 2023).

At different stages of production, effluents are generated with high concentrations of chemical oxygen demand (COD), biochemical oxygen demand (BOD₅), and other compounds that are difficult to degrade due to their recalcitrant nature, such as phenolic compounds and lignin derivatives, which can also color the effluent (Machado et al., 2018). When left untreated, these effluents can compromise the self-purification capacity of receiving watercourses and affect aquatic fauna and flora (Singh et al., 2022). Specifically for unbleached Kraft effluents, Ramos et al. (2022) noted a significant estrogenic activity, which can alter the reproductive physiology of aquatic organisms. It is imperative to highlight that conventional biological treatments systems cannot fully remove recalcitrant compounds, which are responsible for the dark color and chemical complexities of such waters (Kumar & Verma, 2023).

The activated sludge (AS) process is the secondary system most widely used for industrial wastewater treatment, achieving organic matter removal efficiencies of 65–99% for BOD₅ and 25–85% for COD (Morales et al., 2015; Ramos et al., 2022). However, this process is highly sensitive to operational

variations, particularly organic load shocks. It also generates large volumes of sludge which require complex and costly management (Chamorro et al., 2022; Esmaeeli et al., 2023; Morales et al., 2015).

Other biological alternatives include moving bed biofilm reactors (MBBRs), which exhibit BOD₅ removal efficiencies between 85 and 95% and COD removal efficiencies between 65 and 75% in pilot systems, though these figures may vary under realistic operating conditions. Its main advantage is the high stability of the biomass attached to moving carriers, although they face limitations regarding the removal of recalcitrant compounds (Patel et al., 2021).

Membrane bioreactors (MBRs) produce high-quality effluents due to the efficient retention of solids and organic compounds with low-molecular weight, achieving COD removal rates between 72.3% and 99% and BOD₅ removal of up to 91.1%, in addition to significant color reduction (55% to 79%) and high efficiency in nutrient and metal removal in advanced systems (Patel et al., 2021). However, they face critical challenges regarding membrane biofouling, which increases operating costs and demands pre-treatment steps. In order to treat such effluents, facultative aerated lagoons are one of the most widely used systems due to their ease of operation and ability to remove approximately 90% of biodegradable organic matter. However, some recalcitrant pollutants, such as lignin derivatives, still remain (Ramos et al., 2022). These systems are preferred in areas with extensive tracts of land, as they require substantial physical space and long hydraulic retention times. Their use is particularly advantageous in warm climates, where high temperatures enhance microbial activity and reduce the residence time required for treatment (Calabrò et al., 2024).

Although advanced techniques such as Advanced Oxidation Processes (AOPs) can remove between 90 and 99% of lignin, and membrane technologies such as nanofiltration can currently achieve COD removal rates exceeding 91%, these methods are generally expensive and impractical for treating the massive volumes of wastewater (up to 300 m³/ton) generated by the pulp and paper industry. In contrast, aerated lagoons have demonstrated technical feasibility for managing flows on a massive industrial scale, offering significantly lower construction and management costs. However, despite these advantages, facultative aerated lagoons present limitations in removing

recalcitrant compounds, such as phenols and lignin derivatives, and can experience transient perturbations under fluctuations in organic loading due to the limited metabolic capacity of the native microbiota. This often leads to the partial degradation of these pollutants and increase in residual effluent color, underscoring the necessity for biological intensification strategies (Esmaeeli et al., 2023; Ramos et al., 2022).

In this context, bioaugmentation is presented as a promising bioremediation technique, in which specific and specialized microorganisms are introduced to degrade compounds of interest through ligninolytic enzymatic activity (Kumar et al., 2022), such as *B. thuringiensis*, improving the removal of pollutants in industrial effluents.

The choice of *B. thuringiensis* is grounded in its autochthonous nature, which favors adaptation to the effluent matrix, and its capacity to produce key enzymes involved in the degradation of lignocellulosic compounds (Grgas et al., 2023; Wu et al., 2022). Although microbial consortia may offer broader metabolic pathways, the application of a single autochthonous strain provides greater experimental control, reduces operational complexity, and offers safer handling conditions for environmental applications (Wu et al., 2022).

In this context, Nunes et al. (2022) identified species of the genus *Bacillus*, including *B. thuringiensis*, in full-scale Kraft effluent treatment systems, suggesting their potential for bioaugmentation strategies. Nevertheless, the specific role of *B. thuringiensis* in the degradation of recalcitrant compounds remains underexplored.

Despite this study focusing on this specific strain, the literature documents the application of diverse microorganisms and consortia for the treatment of these effluents. Recent studies employing bacterial consortia comprising *Bacillus* and *Acinetobacter* in pulp and paper wastewater treatment have demonstrated high removal efficiencies at laboratory scale, achieving up to 94% Kraft lignin, 86% color, 91% COD, and 97% total phenolic compounds under controlled conditions (Kumar & Verma, 2025). Gupta and Kapoor (2024) show that the use of different autochthonous bacterial consortia, i.e., isolated from the same matrix in the Kraft wastewater treatment plant, promotes COD removals between 70 to 80%, with color and lignin reduction between 54 and

97%, respectively. Furthermore, other species such as *Bacillus cereus*, *Acinetobacter baumannii*, and *Pseudomonas putida* have also demonstrated high removal rates of COD, phenolic compounds, and lignin derivatives (Haq & Kalamdhad, 2023; Kumar et al., 2022; Verma & Chandra, 2024). These approaches offer distinct advantages, such as higher metabolic diversity and significant removal of recalcitrant compounds; however, most of these studies are conducted under controlled, laboratory-scale conditions, limiting their direct application in configurations like facultative aerated lagoons. In practical or full-scale scenarios, continuous bioaugmentation does not always yield substantial performance enhancements, exhibiting limited efficacy due to operational complexity and competition with the native microbiota (Mattingly et al., 2025). To overcome these operational constraints, bioaugmentation can be enhanced by employing immobilized microorganisms on support media, as biofilm formation increases cell retention, enhances process tolerance to hydraulic and organic load fluctuations, and reduces sludge production (Chaudhry & Paliwal, 2018).

Commercial products containing cereal bran have been used as a support medium for bacteria in bioremediation processes. In this sense, ground rice husk is an interesting alternative, as it is an abundant and accessible agro-industrial residue that provides a large surface area for the formation of bacterial biofilms (Lazzaretti, 2001; Saraiva, 2025). The novelty of this study lies in applying *B. thuringiensis* immobilized on this residue directly within facultative aerated lagoons as a complementary biological intensification strategy, aiming not to replace the conventional system, but rather to enhance its performance. Compared to non-bioaugmented processes, which experience efficiency losses under operational fluctuations, this approach accelerates system recovery following organic shock loads, offering a sustainable, cost-effective, and promising strategy for Kraft mill wastewater. Despite the wide variety of available technologies, there is still a lack of studies evaluating strategies to improve the removal of recalcitrant compounds in conventional lagoon systems under dynamic operating conditions, such as shock loads. In particular, the integration of bioaugmentation using immobilized autochthonous microorganisms in facultative aerated lagoons across different operational scales remains scarcely explored. Therefore, this study aims to

address this gap by proposing a combined approach that enhances treatment performance while maintaining the operational advantages of lagoon systems.

The objective of this research is to evaluate the performance and operational response of facultative aerated lagoons (FAL) at bench and pilot scale subjected to load shocks, using bioaugmentation with *Bacillus thuringiensis* NR114581.1 immobilized on rice husks.

2 Methods

2.1 Sampling the Kraft Effluent

The unbleached Kraft pulp effluent samples used in this research were collected at the inlet of the wastewater treatment plant of a pulp mill located in the Metropolitan Region of Curitiba, Brazil, (approximate coordinates: 25°35'S, 49°24'W), using the grab sampling technique. The samples were transported in 20 L drums and kept refrigerated at 4 °C. Additionally, a sample was collected from the pressing process to prepare composite samples (1:1 v/v) to feed the bench FAL and to also increase the organic matter content when applying load shocks, simulating possible leaks from the industrial process. The physicochemical parameters characterized included chemical oxygen demand (COD), which was determined by the closed reflux colorimetric method (Method 5220 D; APHA, 2023) with absorbance readings taken on a UV–Vis spectrophotometer (Cary 50 Conc, Varian Inc., Palo Alto, USA), and a five-day biochemical oxygen demand (BOD₅), which was determined by the dilution method (Method 5210 B), with dissolved oxygen measured using the azide modification of the Winkler method (Method 4500-O C; APHA, 2023). The total phenolic compounds (TPC), lignin compounds (LC), lignosulfonic compounds (LSC), aromatic compounds (AC), and color were determined by UV–Vis spectrophotometry using the same equipment, following adaptations of the methodology proposed by Melchiori et al. (2024) and Chamorro et al. (2022). Turbidity was measured using a microprocessor-controlled digital turbidimeter (DeLAB, model Dlt-wv, Araraquara, Brazil).

Samples were pre-filtered through 0.45 µm cellulose membranes for the physicochemical analyses, except for turbidity determination. All characterization analyses

were performed in triplicate. Raw effluent was used to feed the reactors and in batch experiments in order to preserve the original characteristics of the matrix.

2.2 Evaluation of Bioaugmentation Under Batch Conditions

B. thuringiensis NR114581.1 was originally isolated from the facultative aerated lagoon in the effluent treatment plant where the effluents were collected, and its molecular identification was carried out through sequencing of the 16S rDNA gene by Nunes et al. (2022). The bacterial immobilization was performed by physical adsorption. Thereafter, rice husks were previously ground and sieved through 20 and 48 mesh screens, selecting the fraction retained on the 48 mesh as the support material, which was subsequently autoclaved. A bacterial culture with approximately 1×10^8 CFU·mL⁻¹, estimated by the McFarland scale, was sprayed onto the rice husks at a proportion of 150 mL of the culture per 100 g of husks (Chaudhry & Paliwal, 2018). Subsequently, the material was dried in a microbiological incubator at 28 °C for 3 days and stored at room temperature. The bacterial concentration associated with the rice husks was estimated by serial dilution and plate counting, obtaining 1×10^{10} CFU·g⁻¹ (Horak et al., 2013).

Batch bioaugmentation tests were performed on 100 mL of effluent samples, applying Central Composite Design (CCD), within the Response Surface Methodology (RSM) to optimize the dosage of bioproduct used in the FAL. The evaluated independent variables were bioproduct concentration (X_1), at levels of 0.2, 0.5 and 1.0 g·L⁻¹, (corresponding to groups A, B and C, respectively) and hydraulic retention time (HRT) (X_2), at levels of 2, 5 and 7 days. The experiment comprised a total of 11 experimental runs, including 4 factorial points, 4 axial points, and 3 central point replicates defined at 0.5 g·L⁻¹ of bioproduct and HRT of 5 days. The monitored response variables were the removals of COD, TPC, LC, AC, LSC and color. The initial volumetric organic loading rate (OLR) selected was 0.2 kg COD m⁻³·d⁻¹, falling within the range reported for bench-scale FAL systems (Nunes et al., 2022). HRTs of 2, 5 and 7 days were selected based on the conditions reported by Mattingly et al. (2025), who presented HRTs of approximately 2–3 days for individual aerated lagoons and 6.2 days for the complete lagoon system in wastewater. Bioproduct concentrations were

established as a preliminary experimental range to evaluate different application levels and select suitable conditions for subsequent continuous trials. This approach accounted for uncertainties regarding commercially applied dosages and the limitations in transferring bioaugmentation results from laboratory scale to larger systems (Mattingly et al., 2025). Additionally, control treatments consisting of raw effluent, effluent with biomass sludge, and effluent with support material (rice husks) were evaluated at an HRT of 7 days for comparative analysis.

The model evaluation considered statistical significance ($p < 0.05$), the absence of significant lack of fit ($p > 0.05$), and the coefficient of determination ($R^2 \geq 0.80$). For responses meeting these statistical criteria (TPC, LC and color), the data were individually fitted according to Eq. 1.

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{12} X_1 X_2 \quad (1)$$

where $\hat{Y}$ represents the predicted response, β_0 is the intercept, β_1 e β_2 are the linear coefficients, β_{11} and β_{22} are the quadratic coefficients, and β_{12} is the interaction coefficient. Statistical analysis and simultaneous optimization were performed using Minitab 21.1 software (Minitab LLC, USA) using the composite

desirability function ranging from 0 to 1, in which values close to 1 indicate the best operating conditions (Leyva et al., 2022).

SEM was applied to rice husks previously lyophilized using a lyophilizer (Liotop L101, Liobrás, São Carlos, Brazil) and metallized using a sputter coater (Quorum Q150R ES, Quorum Technologies, Lewes, UK), under inoculum-free conditions, after immobilization and batch treatments, with magnifications 2000x, 3000x, 5000x, and 10000x, at the Multi-User Center for Materials Characterization from UTFPR.

2.3 Operation and Performance of the Bench and Pilot-scale FAL

A 1L bench-scale FAL was operated for 219 days with HRT of 3.29 ± 0.99 d (Fig. 1), which exhibited variations resulting from flow rate adjustments tied to maintenance and deliberate modifications of the volumetric organic loading rate (OLR) across different operating conditions of the system. The baseline OLR applied to the system was $0.2 \text{ kg COD m}^{-3} \cdot \text{d}^{-1}$, as determined during the batch optimization stage. The nutrient ratio used was 100:0.5:0.1 for COD:N:P, following the methodology described by Pazda and Xavier (2026). Nutrient supplementation

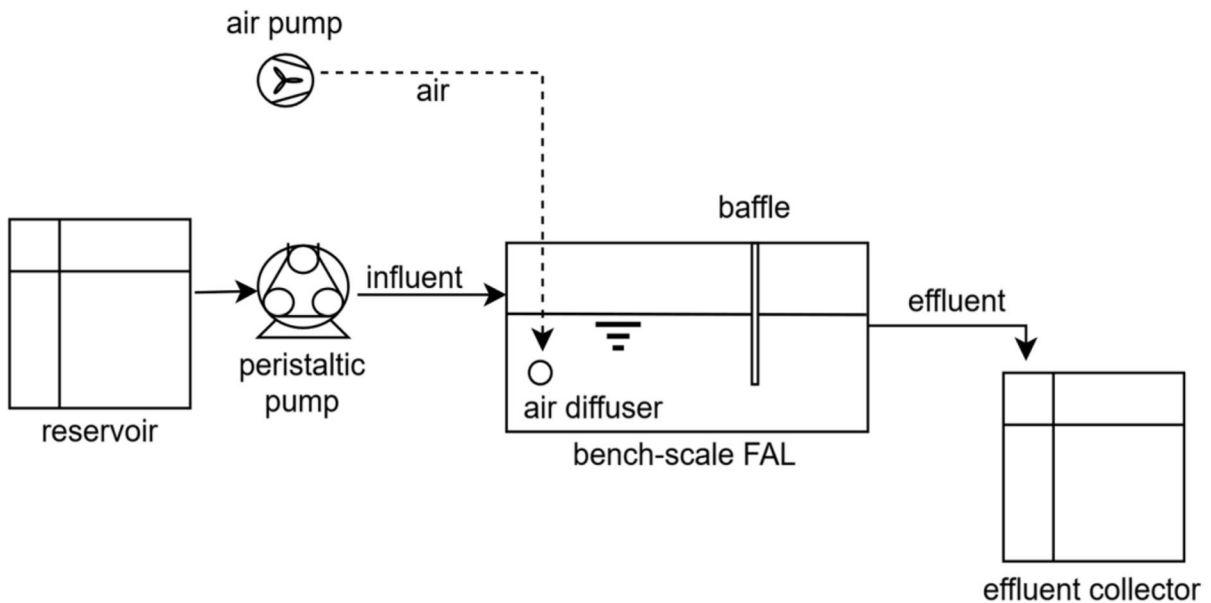


Fig. 1 Schematic diagram of the bench-scale FAL experimental setup (1 L). The effluent is pumped from the reservoir using a peristaltic pump; air supply is provided by an air pump and

distributed through a porous diffuser at the reactor inlet. The system features an internal baffle to assist in suspended solids settling and a container for treated effluent storage

was performed using ammonium chloride (NH_4Cl) as the nitrogen source and dipotassium phosphate (K_2HPO_4) as the phosphorus source. The pH of the influent was adjusted to 7.05 ± 0.03 and monitored using a digital pH meter (Incoterm, model pH DLT-WV, Porto Alegre, Brazil). The system was inoculated with biomass ($70 \text{ mg VSS} \cdot \text{L}^{-1}$) and continuously aerated by means of a dual-outlet air pump (BOYU, model SC-7500, Guangdong, China), operating at a total flow rate of $6 \text{ L} \cdot \text{min}^{-1}$. The airflow was distributed in the system using a $15 \times 25 \text{ mm}$ cylindrical diffuser stone strategically positioned at the flow inlet. The dissolved oxygen concentration was maintained at $6.22 \pm 0.35 \text{ mg O}_2 \cdot \text{L}^{-1}$, measured with a digital oximeter (Instrutherm, model MO-920, São Paulo, Brazil). The influent was supplied by a peristaltic pump (Model BP 600, Milan, Colombo, Brazil) and 4 treatment steps were evaluated: i) stabilization of the FAL; ii) shock application with a 1.8-fold increase in the OLR baseline; iii) shock application with a sixfold increase in the OLR baseline; iv) shock application with a sixfold increase in the OLR baseline followed by bioaugmentation. The shocks were applied for 48 h, new shocks were administered every 7 days, and this procedure was repeated 3 times. The bioaugmentation was performed with a single application at the beginning of the shock, using rice husks ($0.2 \text{ g} \cdot \text{L}^{-1}$) containing *B. thuringiensis* NR114581.1, $1 \times 10^{10} \text{ CFU} \cdot \text{g}^{-1}$. The non-bioaugmented shock evaluation comprised an initial exploratory trial at

1.8 times the baseline OLR, followed by two shock cycles at 6 times the baseline OLR (operational duplicate). In turn, the sixfold OLR shock stage combined with bioaugmentation was conducted in operational triplicate. All analytical determinations of the collected effluent were performed in triplicate.

The 1 m^3 pilot-scale FAL was scaled up 1000 times compared to the bench-scale FAL volume (Fig. 2), while maintaining the nutrient-to-biomass inoculum ratio. For this larger scale, nutrient adjustment was performed using the same products employed in the industrial wastewater treatment plants: commercial urea (as a nitrogen source) and monoammonium phosphate (as a phosphorus source). Oxygen was supplied to the system through two air pumps of the same model used at the bench scale, operating at a total flow rate of $6 \text{ L} \cdot \text{min}^{-1}$. The airflow was distributed through four $15 \times 25 \text{ mm}$ diffuser stones positioned at the reactor inlet at mid-depth. Under these conditions, the dissolved oxygen concentration remained at $3.36 \pm 1.48 \text{ mg O}_2 \cdot \text{L}^{-1}$, monitored using the same oximeter employed for the bench-scale setup. These values remained above $2.0 \text{ mg O}_2 \cdot \text{L}^{-1}$, a range generally considered adequate for aerobic biological processes (Chamorro & Vidal, 2015). The pilot system was operated for 112 days and installed near the industrial treatment plant, gravity-fed with effluent from the primary settling pond of the plant. The baseline OLR was $0.1 \text{ kg COD m}^{-3} \cdot \text{d}^{-1}$ due to the low contaminant load present in the industrial effluent during the study period. The HRT was

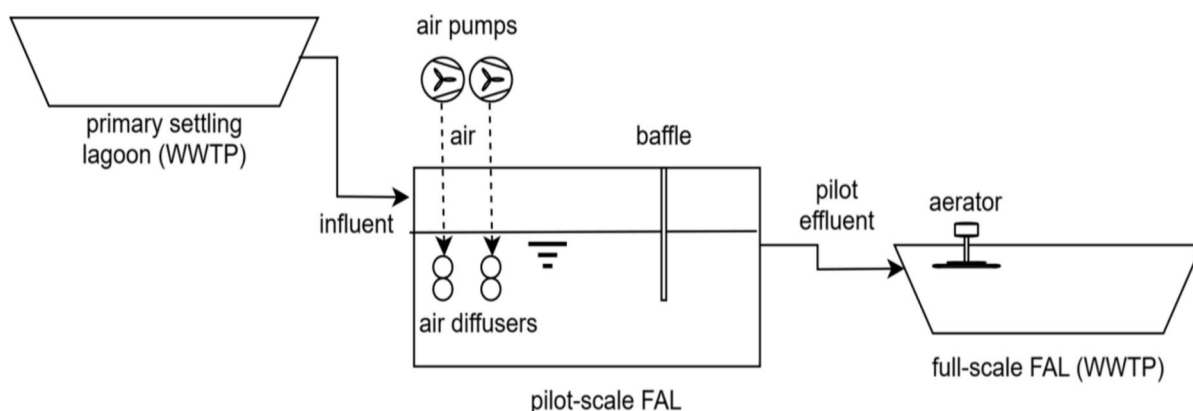


Fig. 2 Schematic representation of the pilot-scale facultative aerated lagoon (FAL) system (working volume of 1 m^3). The system is gravity-fed with effluent from the industrial WWTP primary settling lagoon. Aeration is provided by two air pumps

connected to four diffusers. The reactor features an internal baffle to assist in solids settling, and the treated effluent is discharged directly into the full-scale FAL

between 2.45 ± 0.24 d as a result of the hydraulic conditions imposed by the gravity-feed configuration of the system, with a pH of 7.00 ± 0.24 . Three steps were evaluated: i) stabilization of the FAL; ii) application of a shock 5 times the baseline OLR; iii) application of a shock 5 times the baseline OLR followed by bioaugmentation. The shocks were applied for 4 h, new shocks were applied every 7 days, and this procedure was repeated 3 times. Bioaugmentation was performed with a single application simultaneously with the start of the shock, using rice husks ($0.2 \text{ g} \cdot \text{L}^{-1}$) containing *B. thuringiensis* NR114581.1, $1 \times 10^{10} \text{ CFU} \cdot \text{g}^{-1}$. The procedure was repeated 3 times after each recovery. For each operational cycle, effluent analytical determinations were performed in analytical triplicate.

2.4 Relative Variation of Fluorescent Organic Matter

Fluorescence excitation-emission matrix (FEEM) measurements were performed using a fluorescence spectrophotometer (Cary Eclipse, Varian Inc, Palo Alto, USA). The 220–540 nm range was selected for excitation and 290–550 nm for emission. The data collection interval was 10 nm, the slit width was 10 nm, and the scan rate was $6,000 \text{ nm} \cdot \text{min}^{-1}$ for the construction of contour maps (Faizel et al., 2024). To minimize the inner filter effect, the samples were diluted 20 times in the bench-scale FAL effluents and 10 times in the pilot-scale FAL effluents (Coble et al., 2014; Faizel et al., 2024). Readings were taken at room temperature and without pH adjustment (Coble et al., 2014).

The peak picking technique was used in the analysis of FEEM for the identification and quantification of spectral regions which are characteristic of fluorescent organic matter (Faizel et al., 2024). Based on the peaks reported by Rodriguez et al. (2022) for pulp and paper effluents, the spectral regions considered were closest to the Excitation/Emission wavelengths in A (230/423 nm), T1 (275/350 nm), T2 (230/360 nm), and M (305/415 nm). The FEEM measurements were evaluated under shock conditions, with and without bioproduct, in the bench-scale FAL (2 and 5 days after shock) and in the pilot FAL (days 1 and 3 after shock).

2.5 Microfauna of Bench-scale and Pilot-scale Systems

The microfauna ecological succession was evaluated in both the bench and pilot-scale treatment

systems by identifying and quantifying the main microfaunistic groups. The data was expressed as arithmetic means per functional group, without applying inferential statistics. Sludge samples were observed under a microscope (20 $\times$ and 40 $\times$) and prepared on unfixed slides, with identification to the genus level using illustrated keys (Finlay et al., 1988; Patterson et al., 1996; Serrano et al., 2008) and sludge microbiology references (Chamorro & Vidal, 2015; Gerardi, 2015; Jenkins et al., 2003). The taxonomic identification of the different microorganisms was based strictly on diagnostic structural and morphological criteria as well as on the locomotion behavior observed in the samples, as established in the cited references. The sampling was carried out in specific operational zones, and the abundance index was estimated on an ordinal scale (1–5), representing: 1 (rare), 2 (uncommon), 3 (frequent), 4 (common), and 5 (very abundant), organized into functional groups (sessile, free-living, and crawling ciliates, flagellates, amoebas, metazoans, and “Others” in the pilot scale) (Oliveira et al., 2009). In the bench-scale FAL, the following steps were considered: Initial (~40 days), Intermediate (~70 days), Mature (~90 days), Shock (~160–167 days), Pre-bioaugmentation (~169–172 days), Post-bioaugmentation (~210 days). In the pilot-scale, two steps were analyzed: Pre-shock (~65 days), and Post-shock (~112 days) with bioaugmentation.

3 Results and Discussion

3.1 Kraft Influent Sample Characterization

Table 1 presents the characterization of the physico-chemical parameters of the 1:1 v/v composite samples used as influents in the bench tests.

Samples 3, and 4 showed BOD₅/COD ratios indicative of a major biodegradable fraction, while samples 1 and 2 showed lower biodegradability. All four samples demonstrated viability to be treated by biological processes (Von Sperling, 2014). The TPC values obtained in this study remained within the range observed by Peitz and Xavier (2020) for samples from the same industrial matrix, between 233 and 397 $\text{mg} \cdot \text{L}^{-1}$. Regarding spectrophotometric characterization, the absorbance levels obtained for AC (254 nm), LC (280 nm), and color (440 nm)

Table 1 Physicochemical characterization of composite Kraft effluent samples. Chemical oxygen demand (COD); biochemical oxygen demand (BOD₅); BOD₅/COD ratio; total phenolic compounds (TPC); lignin compounds (LC); lignosulfonic compounds (LSC); aromatic compounds (AC) and color

Parameter	Sample 1	Sample 2	Sample 3	Sample 4
COD (mg·L ⁻¹)	679.20 ± 9.07	874.72 ± 1.29	401.34 ± 4.28	906.24 ± 10.44
DBO ₅ (mg·L ⁻¹)	289.50 ± 6.36	282.00 ± 8.49	212.15 ± 3.04	495.00 ± 21.21
DBO ₅ /COD	0.43	0.32	0.53	0.54
TPC (mg·L ⁻¹)	320.77 ± 0.60	278.85 ± 2.84	219.81 ± 2.84	380.85 ± 0.38
LC (1 × 1 cm)	4.21 ± 0.03	4.25 ± 0.05	3.04 ± 0.01	5.94 ± 0.01
LSC (1 × 1 cm)	1.34 ± 0.01	1.43 ± 0.02	1.04 ± 0.01	2.03 ± 0.01
AC (1 × 1 cm)	5.37 ± 0.01	5.88 ± 0.01	4.51 ± 0.01	6.96 ± 0.01
color (1 × 1 cm)	0.39 ± 0.03	0.37 ± 0.02	0.24 ± 0.01	0.46 ± 0.01

are consistent with the ranges reported by Peitz and Xavier (2020) and Nunes et al. (2022) for unbleached Kraft effluents. These studies reported values ranging from 3.191 to 6.285 abs for AC, 2.786 to 6.352 abs for LC, and 0.222 to 0.402 abs for color. These ranges reflect the high load of recalcitrant organic matter which is characteristic of this type of effluent.

In contrast, bleached Kraft effluents exhibit significantly lower absorbance values, as reported by Hidd et al. (2025) and Chamorro et al. (2022), with mean values of 0.59 for AC, 2.12 for LC, and 0.15 for color. This deviation can be attributed to the bleaching process, which reduces the concentration of compounds associated with color and recalcitrance, resulting in an effluent with lower structural complexity and higher biodegradability compared to the unbleached Kraft effluent evaluated in this study.

Furthermore, the combined analysis of the BOD₅/COD ratio, TPC, and UV–Vis absorbance reveals that the effluent contains a significant fraction of persistent organic matter associated with aromatic compounds and lignin derivatives. This characteristic indicates that, although biological treatment is feasible, the presence of these compounds may limit process efficiency, particularly regarding the removal of color and recalcitrant organic matter. Consequently, the results suggest a need for complementary strategies or appropriate operational conditions to optimize the performance of biological treatment applied to this type of effluent (Ramos et al., 2022).

3.2 Batch Bioaugmentation

In the bioaugmentation treatments, composite sample 1 from Table 1 was utilized. Figure 3 below presents

the performance obtained in removing organic matter and specific contaminants throughout the experiment.

The highest removal efficiencies were observed on day 5, particularly in treatment A, which achieved 56% COD removal. These values surpass those reported by Melchioris et al. (2024), who obtained 48% COD removal in an MBBR system with sponges as a support medium for *Bacillus* sp., while TPC removal was 51%. The removal of LC, AC, LSC was 38%, 52%, and 35%, respectively, and the color was reduced by 45%. Nunes et al. (2022), when freely applying *B. thuringiensis* NR114581.1 associated with biomass, reported values of 30% and 50% for LC and AC, LSC removal was less than 20%, and color decreased by only 4%.

Lignin degradation by bacteria typically involves extracellular and intracellular oxidoreductase enzymes and specific catabolic pathways, such as the β-ketoadipate pathway, for aromatic ring cleavage and integration into the tricarboxylic acid (TCA) cycle (Grgas et al., 2023). Specifically, *B. thuringiensis* can produce lignin-modifying enzymes like laccases (Lac), manganese peroxidases (MnP), and lignin peroxidases (LiP), driving the depolymerization of lignin and phenolic compounds (Wu et al., 2022). Although evidence for *B. thuringiensis* in Kraft effluent remains emerging, *Bacillus* species are widely recognized for degrading recalcitrant aromatics through these mechanisms (Kumar et al., 2022; Nunes et al., 2022), supporting the observed transformation of lignin compounds in this system.

In this context, the decrease in COD indicates partial mineralization of organic matter, while the reductions in LC and LSC suggest lignin depolymerization processes. Similarly, the removal of TPC and color aligns with the transformation of phenolic and chromophoric structures. The higher removal efficiencies observed in this study compared to previous

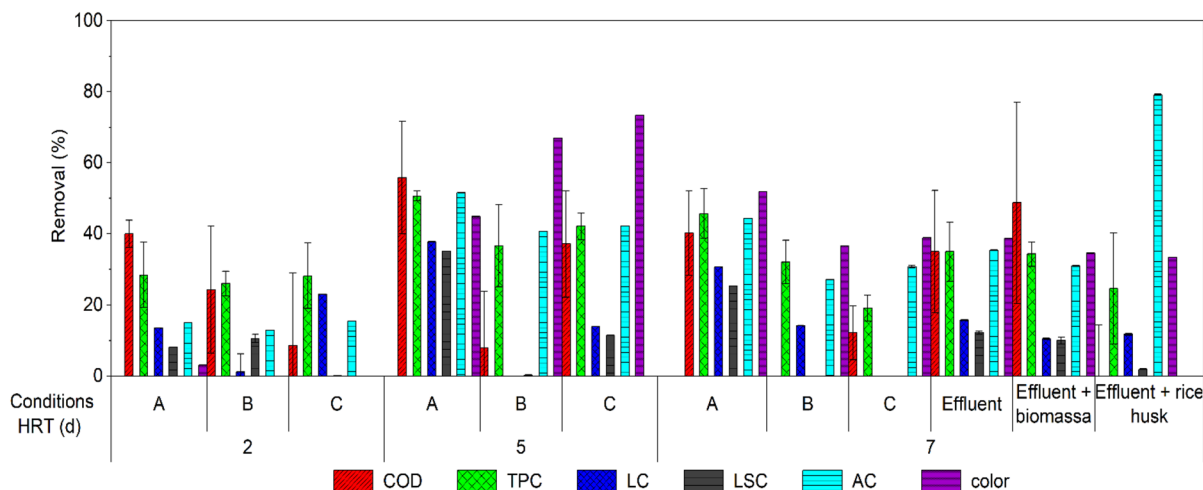


Fig. 3 Batch bioaugmentation tests under three conditions as a function of the bioproduct concentration (*B. thuringiensis* immobilized on rice husks): A ($0.2 \text{ g}\cdot\text{L}^{-1}$), B ($0.5 \text{ g}\cdot\text{L}^{-1}$) and C ($1.0 \text{ g}\cdot\text{L}^{-1}$). The removal of chemical oxygen demand (COD), total phenolic compounds (TPC), lignin compounds (LC), lignosulfonic compounds (LSC), aromatic compounds (AC)

and color was evaluated at hydraulic retention times (HRT) of 2, 5 and 7 days. The controls consisted of Composite Sample 1 (effluent), Composite Sample 1 (effluent) with biomass sludge, and Composite Sample 1 (effluent) with rice husks. Control treatments were evaluated on day 7

reports may be attributed to the specific operational characteristics of the evaluated biological system.

From day 5 onwards, a decrease in the efficiency of compound removal can be observed, possibly due to the partial recombination of macromolecules, forming phenolic compounds and chromophore intermediates (Peitz & Xavier, 2020). This behavior is consistent with the hypothesis of secondary phenolic metabolite formation and accumulation over time (Melchior et al., 2024). Members of the genus *Bacillus* have been identified as autochthonous microorganisms in Kraft effluent treatment systems (Nunes et al., 2022), which may contribute to the observed performance. Although the efficiency of bioaugmentation processes depends on multiple operational and environmental factors such as pH, nutrient availability, and oxygenation conditions (Mattingly et al., 2025), this study focused on bioproduct dosage and hydraulic retention time as key variables. In this context, the application of response surface methodology (RSM) becomes relevant to evaluate the combined effects of these variables and identify optimal operating conditions in a complex biological system (Ajao et al., 2024).

Statistical analysis via RSM confirmed significant quadratic models for TPC, LC, and color ($R^2=93.68\%$, 91.86% , and 85.88% ; $p=0.005$, 0.009 , and 0.035 , respectively), with no significant lack of

fit ($p>0.05$). The resulting models are expressed in Eq. 2, 3 and 4.

$$\begin{aligned} \text{TPC} = & -130.1 + 5.0X_1 + 73.3X_2 + 1.2X_1^2 \\ & - 6.97X_2^2 - 0.59X_1X_2 \end{aligned} \quad (2)$$

$$\begin{aligned} \text{LC} = & 0.61 - 38.9X_1 + 22.31X_2 + 44.7X_1^2 \\ & - 1.97X_2^2 - 6.38X_1X_2 \end{aligned} \quad (3)$$

$$\begin{aligned} \text{color} = & 28.5 - 131.1X_1 + 6.29X_2 \\ & + 132.1X_1^2 - 0.011X_2^2 - 9.63X_1X_2 \end{aligned} \quad (4)$$

where X_1 represents the bioproduct concentration ($\text{g}\cdot\text{L}^{-1}$) and X_2 represents the HRT (days). The remaining responses were not significant ($p>0.05$) or showed lack of fit and were excluded from optimization. The estimated optimal conditions corresponded to a dosage of approximately $0.2 \text{ g}\cdot\text{L}^{-1}$ and an HRT of 5 days, with a composite desirability function of 0.903, indicating a favorable compromise between the evaluated responses (Ajao et al., 2024).

Regarding controls, the composite sample containing only biomass showed a 55% removal of COD and approximately 30% for the other compounds, while the treatment exclusively with husks did not show

removal of COD and LSC, but there was a decrease in AC of approximately 80%. Figure 4 shows the results of the SEM analyses of the rice husks.

Images 4A (2000x) and 4B (10000x) show that the rice husks surface without bacteria is predominantly smooth, with fissures and sparse adhered solids, possibly related to impurities in the material, without evidence of microbial presence. While in images 4C (3000x) and 4D (10000x), a high density of bacillary structures compatible with the *Bacillus* group can be observed, adhered in an organized manner, in addition to aggregated formations and masses adhered to the surface. These characteristics were also observed by Saraiva (2025) when using rice husks as a ligno-cellulosic support for the dispersion of microorganisms in biological reactors in wastewater treatment.

Following batch treatment, images 4E (2000x) and 4F (5000x) reveal dense, extensive biofilm development across the support surface. This behavior aligns with studies demonstrating that the porous structure of rice husks promotes stable biofilm formation by autochthonous genera (including *Arthrobacter*, *Bacillus*, *Rhodococcus*, and *Staphylococcus*), mediated by extracellular polymeric substances (EPS) for biomass attachment (Alitalieshi et al., 2024). Furthermore, multi-species consortia such as *Bacillus licheniformis*, *Microbacterium* sp., and *Brevundimonas* sp. significantly intensify cell aggregation and biofilm stability under stress conditions via enhanced EPS production (Farias et al., 2025). Although the system was initially inoculated with *B. thuringiensis*, native effluent microorganisms likely co-colonized the support, contributing to this complex biofilm architecture.

In images G (2000x) and H (5000x), possible mineral structures are shown; the crystal formation can be attributed to an increase in the system's pH or to microbial activity, both mechanisms being capable of inducing conditions of supersaturation and precipitation of CaCO_3 . Specifically, Wang et al. (2025) stated that alkaline environments shift the carbon equilibrium to favor carbonate formation, whereas Ferral et al. (2020) showed that *Bacillus* species promote precipitation via a non-ureolytic pathway, using extracellular polymeric substances (EPS) as calcium nucleation sites. In this study, this suggests that the heterotrophic biofilm on the rice husks may have created the microenvironmental conditions for the crystalline arrangements observed via SEM.

3.3 Bench-Scale FAL Performance and Operation

In the stabilization stage, the average OLR during the bench-scale FAL operation was $0.25 \pm 0.19 \text{ kg COD} \cdot \text{m}^{-3} \cdot \text{d}^{-1}$, with an average HRT of $3.29 \pm 0.99 \text{ d}$. Such results are similar to those reported by Coelho et al. (2024), who recorded $0.2 \text{ kg COD} \cdot \text{m}^{-3} \cdot \text{d}^{-1}$ and an HRT of 3.78 d in a 1 L FAL reactor treating Kraft effluents for 64 days. There was an increase in pH to 7.95 ± 0.26 , possibly related to the formation of carbonates in the sedimentation zone of the facultative lagoon systems. This observation corroborates the presence of minerals evidenced by SEM of rice husks after biological treatment (Von Sperling, 2017). The removal of organic matter, specific compounds, color, and turbidity are detailed in Fig. 5.

The bench-scale system showed removals of 47% for COD and 55% for BOD_5 during the stabilization stage. Coelho et al. (2024), reported removals of 36% for COD and 73% for BOD_5 when treating pulp mill effluents in a FAL operated with an HRT of 3.78 d and a COD of $0.2 \text{ kg} \cdot \text{m}^{-3} \cdot \text{d}^{-1}$. During organic shocks, the COD exhibited increased variability, especially in condition C, highlighting the system's sensitivity to high loads and the need for recovery time. Similar behavior was observed by Morales et al. (2015), where black liquor shocks (up to 3.225 mg COD/L) applied for 24 h resulted in an initial removal peak (76%) followed by a drastic drop (−19%) after two days, due to the retention of recalcitrant matter in the system. Likewise, Chamorro et al. (2022) observed a progressive reduction in COD removal efficiency (from 64.84% to 20.34%) when reducing the HRT from 24 to 12 h during the shocks; this increased the organic load and momentarily destabilized heterotrophic biomass activity. These findings corroborate the metabolic instability and the need for a biomass recovery period observed in the present study when facing high loads and varying exposure times.

During the bioaugmentation, an average reduction in COD of 36% was observed, but with a lower median efficiency (33%), indicating partial recovery of the process. In contrast, BOD_5 showed more robust behavior, with a progressive increase in efficiency, reaching 72% in the bioaugmentation shock, suggesting greater resilience of the system for the removal of the biodegradable fraction of organic matter compared to the recalcitrant fraction. During the bioaugmentation step, the observed effects should be

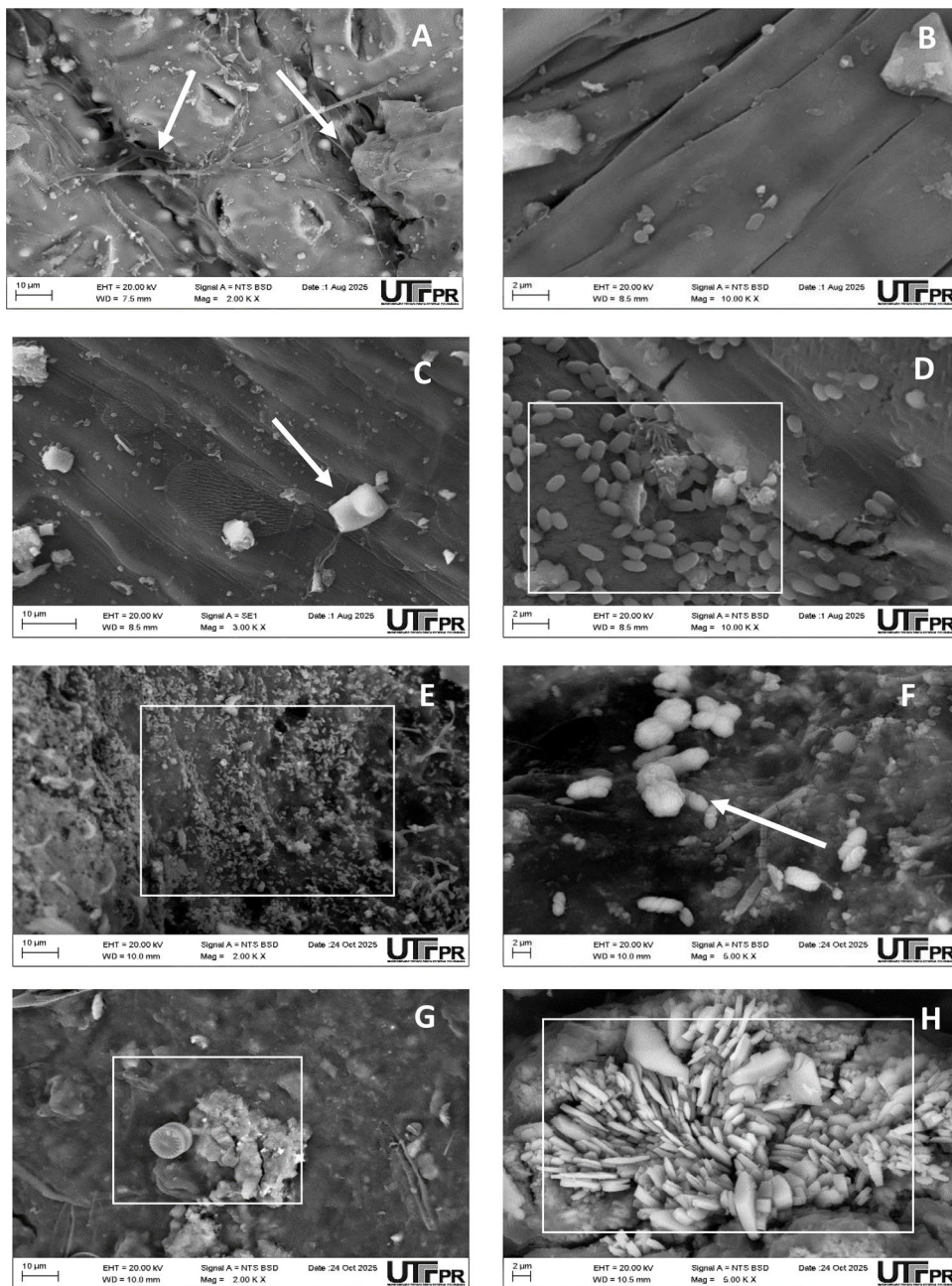
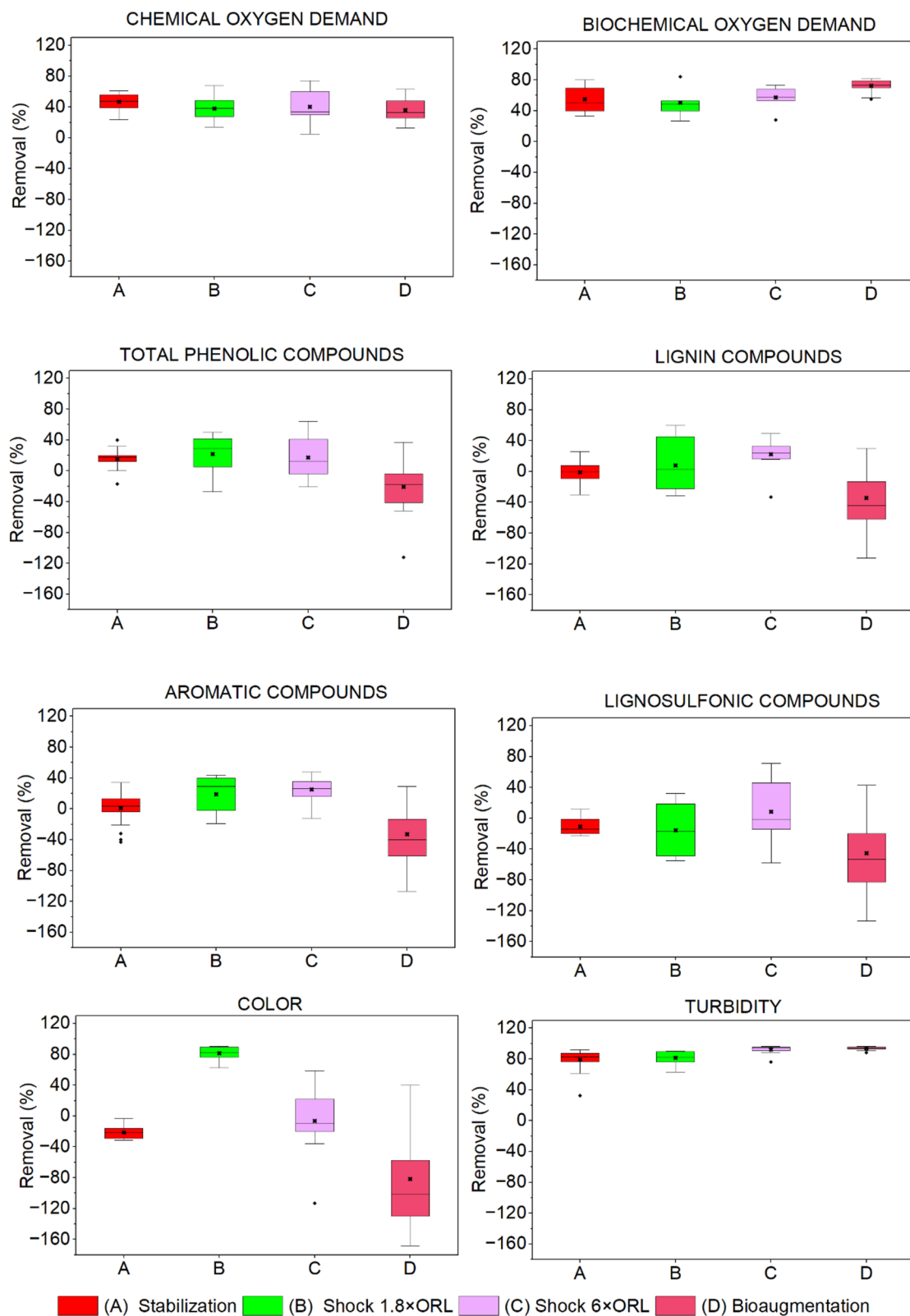


Fig. 4 Scanning electron microscopy (SEM) of rice husks before and after bacterial immobilization and biological treatment. **A** Rice husks without bacteria, 2000× magnification. **B** Rice husks without bacteria, 10000× magnification. **C** Rice husks with immobilized bacteria, 3000× magnification. **D** Rice husks with immobilized bacteria, 10000× magnification.

E Biofilm formation on the husks after biological treatment, 2000× magnification. **F** Biofilm on the husks after treatment, 5000× magnification. **G** Mineral structures on the husks after biological treatment, 2000× magnification. **H** Mineral structures on the husks after biological treatment, 5000× magnification.



◀**Fig. 5** Removal of organic matter, specific compounds, color, and turbidity in FAL at bench scale. Each point within the boxplot represents the average of one of the operation steps: A: stabilization; B: shock treatment of 1.8 times the organic loading rate (OLR); C: shock treatment of 6 times the OLR; D: shock treatment of 6 times the OLR followed by bioaugmentation with *B. thuringiensis* immobilized on rice husks. The box boundaries indicate the interval corresponding to the central 50% of the concentration data, and the central line represents the median value. Each plot represents a parameter

interpreted with caution as they may reflect the combined activity of the inoculated microorganisms and the native or adapted microbial community, particularly under dynamic conditions such as organic load shocks.

The TPC and AC compounds showed average removals of 15% and 1%, respectively, while the other specific compounds and color showed low efficiencies during the stability, with medians close to zero. In shocks B and C, high dispersion and performance limitations were observed under higher loads. Bioaugmentation was associated with negative medians and high variability, suggesting a lack of consistent improvement and possible destabilization of the process. Morales et al. (2015) reported a specific increase in efficiency after shocks, followed by a drop in subsequent days. Studies indicate that bioaugmentation may not be effective even with autochthonous bacteria due to operational limitations such as load shocks, pH variations, temperature, and nutrient deficiencies (Raper et al., 2018).

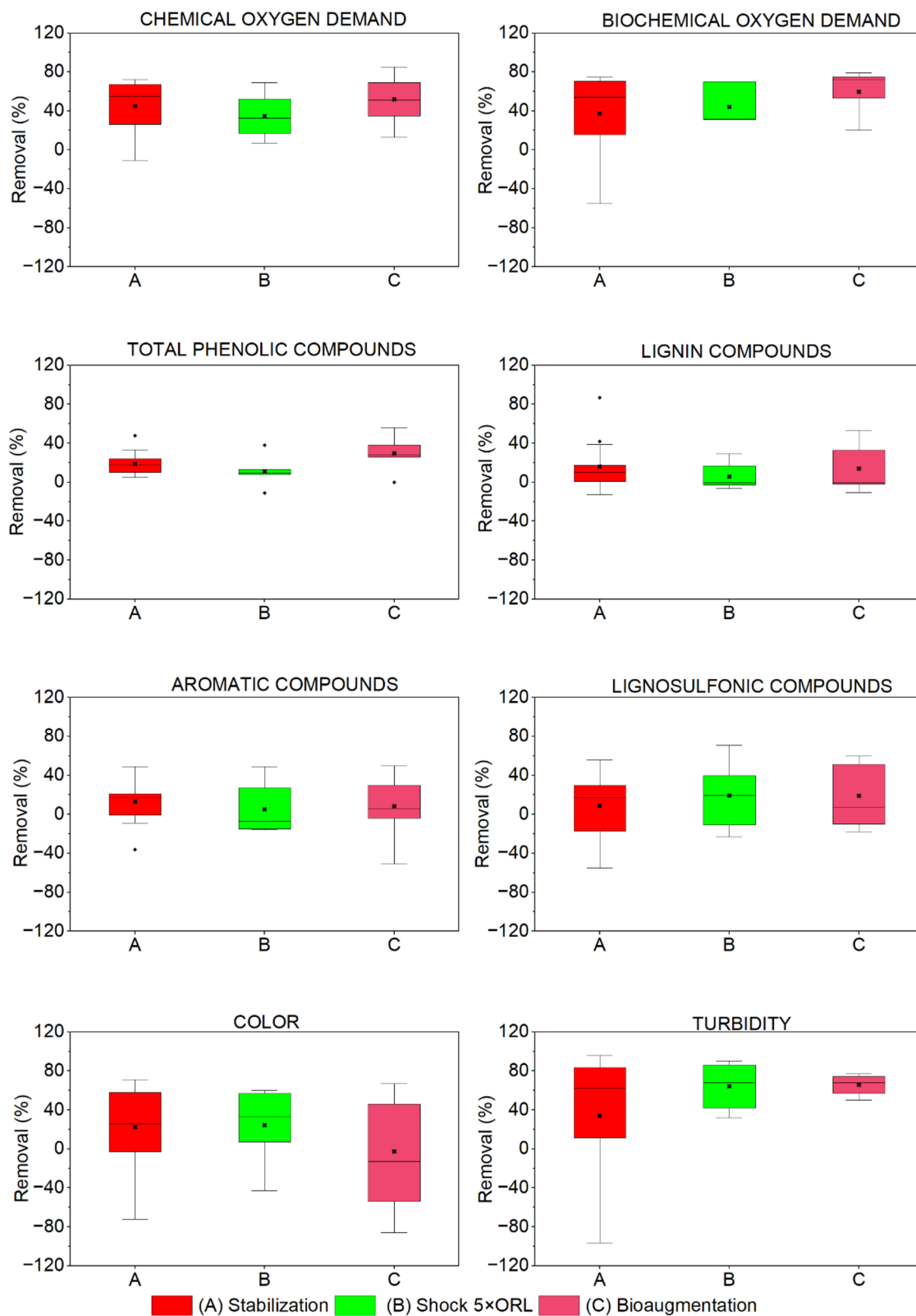
Overall, the evaluated system falls within the category of biological wastewater treatment technologies, such as conventional activated sludge and membrane bioreactors (MBR), although the latter operate under more controlled conditions and achieve higher organic matter removal efficiencies (Patel et al., 2021; Ramos et al., 2022). In contrast, the studied system (operating at the bench scale with a simplified configuration) exhibits greater sensitivity to organic load fluctuations and the presence of recalcitrant compounds, while also involving less operational complexity. Therefore, the obtained results should be interpreted within the context of biological systems applied to the treatment of complex industrial effluents, where process stability and biomass adaptability are key factors.

3.4 Pilot-scale FAL Performance and Operation

The average OLR of this system was $0.10 \pm 0.03 \text{ kg COD} \cdot \text{m}^{-3} \cdot \text{d}^{-1}$, a value closer to that of the industrial-scale FAL ($0.06 \text{ kg COD} \cdot \text{m}^{-3} \cdot \text{d}^{-1}$). The FAL had frequent blockages due to solids in the inlet valve, compromising the flow rate. This limitation also influenced the HRT, with an average of $2.45 \pm 0.47 \text{ d}$. Regarding pH, the influent had an average of 7.00 ± 0.24 , and the effluent 7.73 ± 0.22 ; a behavior similar to that observed in the bench-scale FAL. On the other hand, the system had temperature fluctuations with minimums of 7°C and maximums of 22°C . According to Romero (1999), the optimal temperature for bacterial activity in the treatment is between 25 and 35°C . The removal of organic matter, specific compounds, color, and turbidity is detailed in Fig. 6.

The maximum removal obtained in stage A was 72% for COD and 75% for BOD_5 , results higher than those observed in the full-scale unit, which showed an efficiency of 53% for COD and 60% for BOD_5 during the research. Throughout the operation of the pilot FAL, hydraulic instability episodes and abrupt increases in organic load associated with the operational variability of effluents from pulp and paper industries were recorded, which negatively impacted removal efficiency during specific moments of the research (days 27 and 50 of operation). Hidd et al. (2025) argue that accidental leaks of green, white, and black liquor can reduce COD removal efficiency by up to 50%. Von Sperling (2014) also highlights that treatment performance is affected by abrupt flow variations due to entrainment and biomass loss.

In step B, the average COD removal was 35% and BOD_5 removal was 44%. Morales et al. (2015) and Chamorro et al. (2022) initially observed an increase in such parameters, followed by a decrease in treatment efficiency when applying organic load shocks in bench-scale systems. However, at pilot scale, this behavior was less pronounced, possibly due to the shorter time interval of the shock (4 h) compared to the bench-scale FAL, which was subjected to 48 h shocks. The COD analysis shows that, during bioaugmentation, the average removal was 52% with a median of 51%, a value close to that observed in the stabilization (55%), indicating a tendency for the system to recover after the shock. For BOD_5 , the average removal during bioaugmentation (60%) exceeded that observed in



◀**Fig. 6** Removal of organic matter, specific compounds, color, and turbidity in FAL at pilot scale. Each point within the box-plot represents the average of one of the operation steps: A: stabilization; B: shock treatment of 5 times the organic loading rate (OLR); C: shock treatment of 5 times the organic loading rate (OLR) followed by bioaugmentation with *B. thuringiensis* immobilized on rice husks. The box boundaries indicate the interval corresponding to the central 50% of the concentration data, and the central line represents the median value. Each plot represents a parameter

stabilization (35%), with a relatively concentrated distribution, suggesting more stable performance for the biodegradable fraction of organic matter.

In general, TPC showed high variability throughout the stages even though bioaugmentation recorded the highest average removal efficiency (30%). The results were variable, with no stable recovery after disturbance by organic shocks. For LC, AC, and LSC, high data dispersion was observed, evidencing unstable system behavior. Bioaugmentation did not promote consistent recovery, with average efficiencies of 14%, 8%, and 19%, respectively, similar to or lower than those recorded in stabilization (16%, 13%, and 9%). Regarding color, the bioaugmentation step showed greater dispersion and a lower removal tendency, while the stabilization and shock stages also exhibited variable results. For turbidity, bioaugmentation showed a higher frequency of favorable turbidity removal values, with an average removal of 66%, indicating a positive effect on the recovery of this parameter after disturbance.

In summary, the pilot FAL demonstrated contaminant removal capacity and partial recovery after shocks, reflecting the system's resilience under operating conditions closer to real-world scenarios. However, the effectiveness of bioaugmentation varied according to the parameter evaluated, with no uniform recovery observed for all specific compounds studied.

3.5 Fluorescent Organic Matter Relative Variation

Fluorescence excitation-emission matrices (FEEMs) were applied to samples from the bench-scale FAL under organic shock and bioaugmentation conditions with *B. thuringiensis* immobilized on rice husks. The variations observed under these conditions are evidenced by the fluorescence peaks shown in Fig. 7 and expressed in terms of arbitrary units (a.u.).

FEEM generally presented four characteristic peaks: T1 and T2, associated with protein-like fluorophores and soluble microbial byproducts (Faizel et al., 2024); M, related to biogenic compounds; and A, corresponding to humic matter of terrestrial origin (Rodriguez et al., 2021).

Between 2 (Fig. 7a) and 5 (Fig. 7b) days after the load shock without bioproduct, peaks T1 and T2 showed a relative decrease of approximately 48%, while M and A showed relative reductions of 18% and 13%, respectively, indicating the possible biotransformation and microbial consumption of aromatic and phenolic compounds from lignin (Cawley et al., 2012; Rodriguez et al., 2022; Yamashita et al., 2015).

With bioaugmentation, the peaks showed a reduction on day 2 (Fig. 7c) (77% T1, 84% T2, 56% A and 35% M), highlighting attenuation of protein-like and humic fluorophores. On day 5 (Fig. 7d), a relative increase in peak intensity was observed, attributed to the release of microbial metabolites and the progressive biotransformation of fluorescent compounds (Rodriguez et al., 2021). FEEM was also employed to in a pilot-scale FAL (Fig. 8).

The FEEM in Fig. 8a shows the effect of the organic load shock without bioaugmentation applied for 4 h on the first day, while Fig. 8b shows the effluent three days after the shock. A relative reduction in peaks T1 and T2 was observed, while peaks A and M appear in the system; these may indicate that these fluorophores correspond to specific compounds and microbial metabolites. The fluorescence intensity was lower compared to the bench-scale FAL; therefore, the dilution was half of the one performed for the effluent of the first FAL. This lower intensity may have occurred due to the bench-scale influent composition (1:1 v/v mixture of composite sample from the overall process and cellulose pressing, Table 1), to the operational alternation of the industry, which includes recycled paper scraps with lower lignin content, and to sun exposure, which contributes to the degradation of fluorescent compounds (Yamashita et al., 2015).

The shock application with bioaugmentation (Fig. 8c), showed a relative reduction in peaks T1, T2, and A, indicating a relative decrease in protein-like and humic fluorophores. The appearance of peak M in the effluent (Fig. 8d) suggests microbial activity, reflecting the presence of microbial metabolites and the progressive transformation of soluble

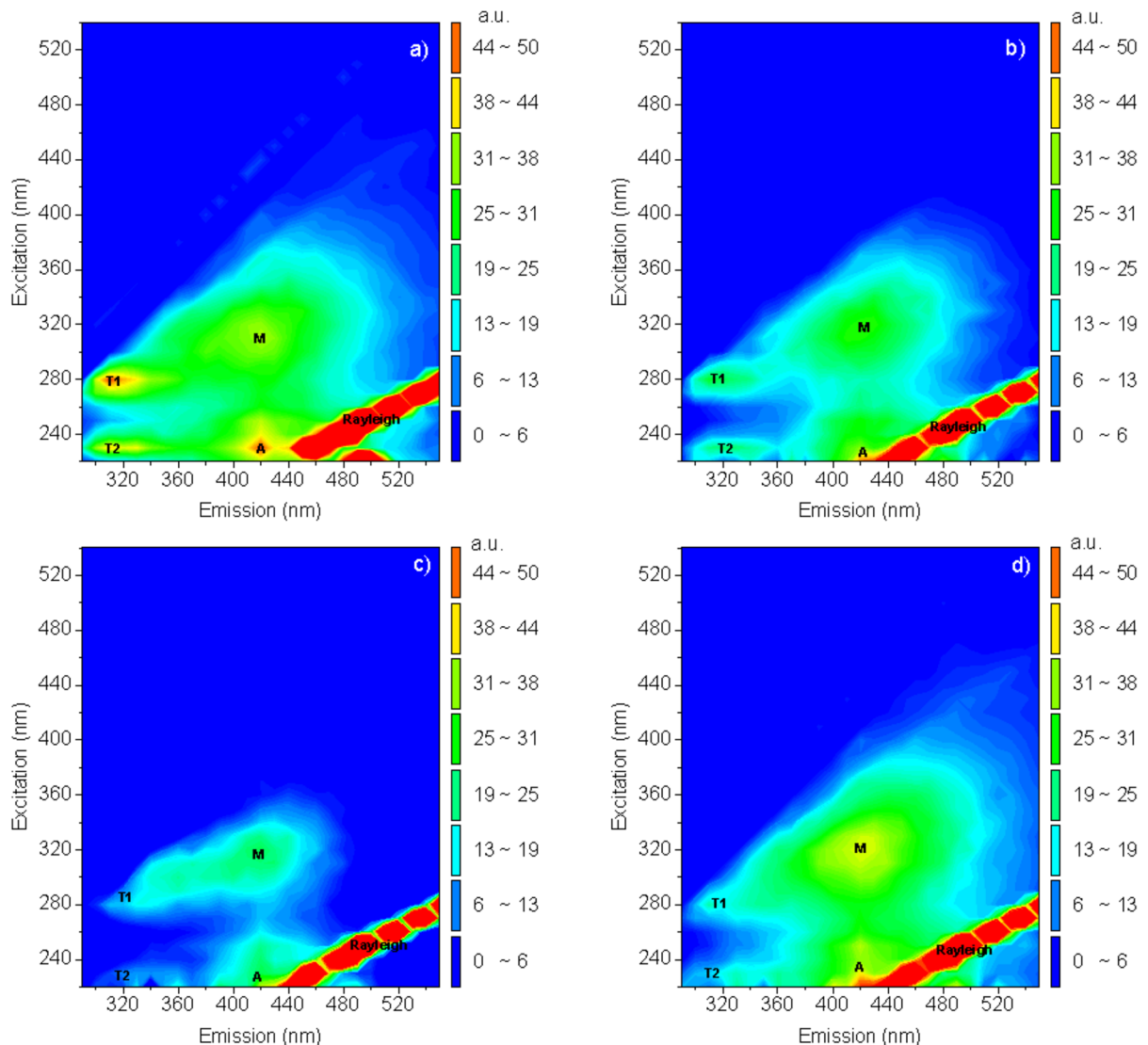


Fig. 7 Fluorescence excitation-emission matrices (FEEMs) applied to bench-scale FAL effluents. Excitation was recorded in the 220–540 nm range (y-axis) and emission in the 290–550 nm range (x-axis), in arbitrary units (a.u.). **a** efflu-

ent 2 days after shock; **b** effluent 5 days after shock; **c** effluent 2 days after shock with bioaugmentation; **d** effluent 5 days after shock with bioaugmentation

organic matter. These data are consistent with the low removal and increase in specific parameters.

3.6 Bench and Pilot-scale Systems Microfauna

The composition of the microfauna structure at bench and pilot scale is presented in Table 2.

Similar microfauna was observed in both systems, with the presence of sessile ciliates, crawling ciliates, metazoans, and filamentous bacteria.

In the pilot system, microalgae and *Euglena* spp. were also identified, characterizing mature sludge with high functional diversity, typical of larger facultative aerated lagoons (Metcalf and Eddy, 2014). Figure 9 shows the average abundance index (on a scale of 1 to 5) of functional groups at bench and pilot scale.

In the bench-scale FAL (Fig. 9A), typical ecological succession was observed throughout the operational period. In the initial phase (40 days), free-living

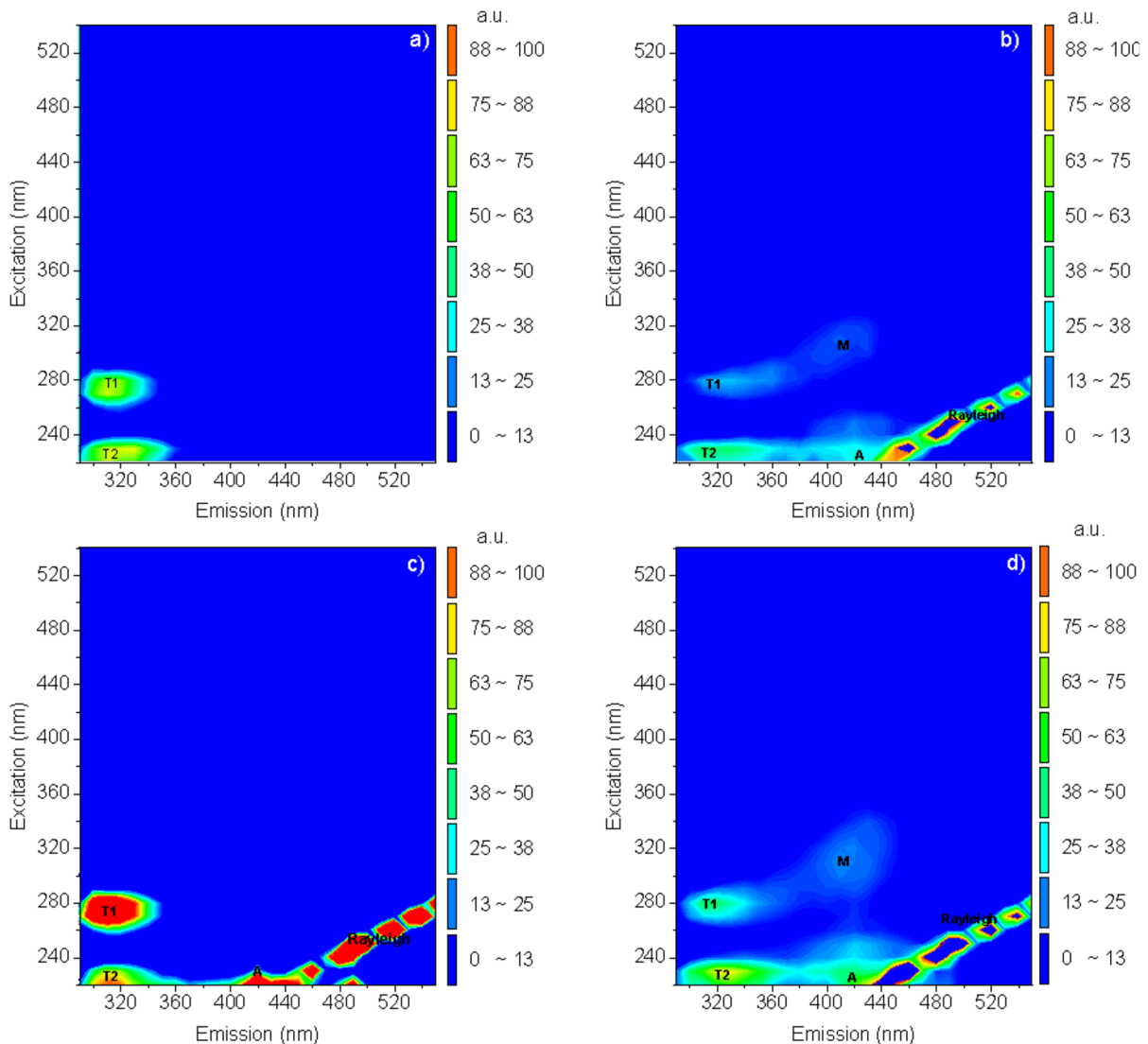


Fig. 8 Fluorescence excitation-emission matrices (FEEMs) applied to FAL effluents at pilot scale. Excitation was recorded in the 220–540 nm range (y-axis) and emission in the 290–550 nm range (x-axis), in arbitrary units (a.u.). **a** effluent day 1

after shock; **b** effluent 3 days after shock; **c** effluent 1 day after shock with bioaugmentation; **d** effluent 3 days after shock with bioaugmentation

ciliates and bacterivorous flagellates predominated, which is characteristic of adapting systems with a high organic load. In this phase, flagellates such as *Bodo saltans* and *Monas* spp., together with free-living ciliates (*Paramecium*, *Tetrahymena*, *Uronema*), act as initial colonizers, consuming dispersed bacteria and promoting initial effluent clarification, which may contribute to COD and BOD removal (Gad et al., 2024; Madoni et al., 2011).

In the intermediate phase (70 days), there was an increase in crawling ciliates, indicating greater sludge stability. Organisms such as *Aspidisca* and *Euplotes* contribute to biological floc organization and improved settleability, favoring the removal of residual organic matter (Eikelboom, 2000; Madoni, 2011). In the mature phase (90 days), the microfauna became dominated by sessile ciliates and metazoans, organisms associated with well-stabilized sludge

Table 2 Microfaunistic composition of the functional groups of sessile ciliates, free ciliates, crawling ciliates, flagellates, amoebas, metazoans and other microorganisms identified at bench and pilot scale

Functional groups	Bench scale	Pilot scale
Sessile ciliates	<i>Vorticella</i> spp., <i>Thuricola folliculata</i> , <i>Opercularia coarctata</i>	<i>Vorticella</i> spp., <i>Thuricola folliculata</i> , <i>Opercularia coarctata</i>
Free ciliates	<i>Colpidium campylum</i> , <i>Paramecium</i> spp., <i>Uronema</i> spp., <i>Tetrahymena</i> spp., <i>Coleps hirtus</i> , <i>Enchelys gasterosteus</i>	<i>Coleps hirtus</i> , <i>Tetrahymena</i> spp., <i>Uronema</i> spp., <i>Paramecium</i> spp., <i>Enchelys gasterosteus</i>
Crawling ciliates	<i>Aspidisca</i> spp., <i>Trithigmostoma</i> spp., <i>Euplotes</i> spp., <i>Stylonychia mytilus</i> , <i>Oxytricha fallax</i>	<i>Aspidisca</i> spp., <i>Trithigmostoma</i> spp., <i>Euplotes</i> spp., <i>Stylonychia mytilus</i> , <i>Oxytricha fallax</i>
Flagellates	<i>Bodo saltans</i> , <i>Monas</i> spp.	<i>Bodo saltans</i> , <i>Peranema trichophorum</i> , <i>Monas</i> spp.
Amoebas	<i>Amoeba proteus</i> , <i>Mayorella</i> spp.	<i>Amoeba proteus</i> , <i>Mayorella</i> spp.
Metazoans	<i>Philodina</i> spp., <i>Rotaria rotatoria</i> , rotifer eggs, <i>Gastrotricha</i> spp.	<i>Philodina</i> spp., <i>Rotaria rotatoria</i> , <i>Brachionus</i> spp., ovos de rotíferos, <i>Gastrotricha</i> spp.
Other microorganisms	-	Filamentous bacteria (likely <i>Microthrix parvicella</i>), microalgae, <i>Euglena</i> spp.

and low residual organic load (Eikelboom, 2000; Metcalf and Eddy, 2014). Sessile ciliates (*Vorticella*, *Opercularia*, *Epistylis*) and metazoans (*Philodina*, *Rotaria*) intensify bacterial predation and contribute to final effluent clarification, in addition to favoring nutrient recycling (Madoni, 2011).

During load shocks, partial ecological regression was observed, with a reduction in sessile ciliates and metazoans and an increase in free-living ciliates and amoebae, typical of unstable environments (Nielsen et al., 2009; Salvadó & Gracia, 1993). The decrease in sessile ciliates and the increase in amoebae reflect a loss of system stability and are commonly associated with adverse operational conditions (Madoni, 2011). After bioaugmentation, an acceleration of ecological succession was observed, with an increase in crawling ciliates and metazoans, indicating recovery of sludge stability. *Aspidisca* spp. and *Philodina* spp. stood out as indicators of good flocculation and process efficiency (Eikelboom, 2000; Jenkins et al., 2003).

At pilot scale (Fig. 9B), the microfauna observed before the shocks showed characteristics which are typical of an ecologically mature system with a high abundance of sessile ciliates, creeping ciliates, metazoans, filamentous bacteria, microalgae, and *Euglena* spp., a pattern which is compatible with larger systems (Metcalf and Eddy, 2014; Nielsen et al., 2009). Microalgae and *Euglena* spp. play a relevant role in system dynamics by contributing to

nutrient assimilation and oxygen supply via photosynthesis, which supports bacterial activity (Gad et al., 2024; Zhang et al., 2023).

After the application of shocks and the bioproduct, a reorganization of the microfaunistic community was observed, with a reduction in flagellates and a relative increase in crawling ciliates, amoebas, and the "Other" group, indicating the maintenance of the structural and functional complexity of the system under disturbance conditions, characteristic of resilient systems (Nielsen et al., 2009). *Aspidisca* spp. and *Oxytricha fallax* stood out among the crawling ciliates, while *Brachionus* spp. was the most abundant metazoan (Eikelboom, 2000; Metcalf and Eddy, 2014). Rotifers (*Philodina*) regulate bacterial populations and contribute to sludge stabilization, reinforcing depuration mechanisms (Madoni, 2011).

The persistent presence of filamentous bacteria, microalgae, and *Euglena* spp. reinforces the facultative nature of the pilot FAL and the ecological resilience of the system, confirming the usefulness of microfauna as an operational diagnostic tool (Eikelboom, 2000; Jenkins et al., 2003). In general, microfauna contributes to clarification, bacterial control, and system stabilization, being associated with improved effluent quality under ecological equilibrium and serving as a robust indicator of operational process status (Gad et al., 2024).

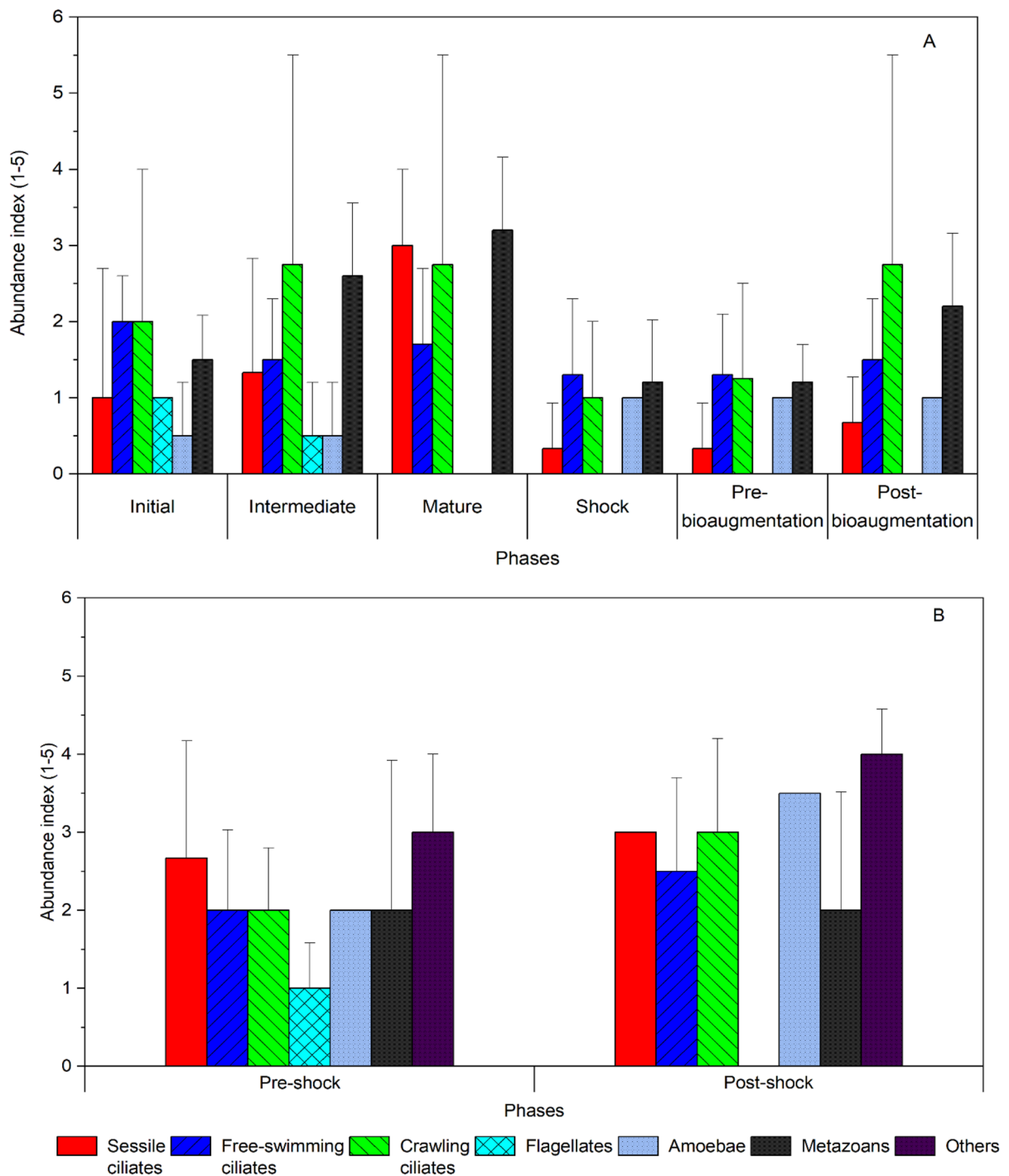


Fig. 9 Composition, temporal dynamics and microfauna succession. **A** Average abundance index in the bench-scale FAL in the stages: Initial (~40 days), Intermediate (~70 days), Mature (~90 days), Shock (~160–167 days), Pre-bioaugmen-

tation (~169–172 days), Post-bioaugmentation (~210 days). **B** Average abundance index in the pilot-scale FAL, Pre-shock (~65 days), Post-shock (~112 days)

4 Conclusions

This study evaluated the performance of facultative aerated lagoons (FAL) subjected to intermittent bioaugmentation with *B. thuringiensis* NR114581.1 immobilized on rice husks under organic shock loads. The results showed that the bioaugmented system exhibited improved removal of biodegradable organic matter (BOD₅ up to 72% at bench-scale and 60% at pilot-scale), while the removal of highly recalcitrant compounds (TPC, LC, AC and LCS) remained limited under continuous dynamic conditions. The pilot-scale system showed greater structural stability, achieving 54% COD removal and more consistent recovery patterns following transient shocks. Immobilization on rice husks was associated with biomass retention and system stability, without compromising the operational simplicity of lagoon systems. Fluorescence spectroscopy (FEEM) and microfaunal analysis (ciliates and rotifers) proved to be useful tools for assessing system stability and biomass dynamics under shock conditions. Overall, the integration of *B. thuringiensis* immobilized on agro-industrial residues represents a feasible and cost-effective strategy to support lagoon performance under variable operational conditions.

Future research should focus on optimizing bioaugmentation dosage strategies and application regimes, as this remains a key gap for improving system performance and enhancing the removal of recalcitrant compounds under continuous-flow conditions.

Acknowledgements The authors would like to thank the Postgraduate Program in Environmental Science and Technology at the Federal Technological University of Paraná (UTFPR), the Academic Master's Program for Innovation (MAI) from the National Council for Scientific and Technological Development (CNPq), the Araucária Foundation, and the Paraná Paper and Pulp Company (COCELPA). The authors also thank the following laboratories at UTFPR–Curitiba: Wastewater Treatment Laboratory (LATAR), Multi-User Laboratory for Chemical Analysis (LAMAQ), Multi-User Laboratory for Equipment and Environmental Analysis (LAMEAA), Laboratory for Research Related to Biomass and Bioenergy (LAPREBB), and the Multi-User Center for Materials Characterization (CMCM).

Author Contributions Jose Antonio Villamizar Florez: Writing – original manuscript version, Research, Formal analysis, Data curation. Esther Soares Assunção Maia: Writing – original manuscript version, Research, Formal analysis, Data curation. Isabela Rita dos Santos Rossi: Writing – original manuscript version, Research, Formal analysis, Data curation. Luiza Dias Silva: Research. Wagner Augusto Rodrigues Julio:

Research. Gustavo Henrique Couto: Supervision and Conceptualization. Claudia Regina Xavier: Writing – revision and editing, Supervision, Resources, Fundraising, Conceptualization.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). Jose Antonio Villamizar Florez, Supported by the Federal Technological University of Paraná (UTFPR). Funded by the Araucária Foundation (CP 17/2023 and CP 23/2023), the National Council for Scientific and Technological Development (CNPq) through the Academic Master's Program for Innovation (MAI) (CP 68/2022), and the Paraná Paper and Pulp Company (COCELPA) (ACT 23064.040994/2024/33).

Esther Soares Assunção Maia, Funded by the National Council for Scientific and Technological Development (CNPq) (MAI) (CP 68/2022) and the Araucária Foundation (CP 17/2023 and CP 23/2023).

Isabela Rita dos Santos Rossi, Funded by the National Council for Scientific and Technological Development (CNPq) (MAI) (CP 68/2022) and the Araucária Foundation (CP 17/2023 and CP 23/2023).

Luiza Silva Dias, Funded by the Araucária Foundation (CP 17/2023 and CP 23/2023).

Data Availability Not applicable.

Declarations

Ethical Approval Not applicable.

Competing interests The authors declare no competing interests.

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